

CS 5814 Project Proposal

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Abstract—This document describes our proposed project: an AI evaluator that predicts the probability that a given AI system answers a question correctly without actually running it. The motivation for this project stems from the need for AI system evaluators that can be trusted in order for organizations such as schools to understand their systems’ capabilities for given tasks. The project will be facilitated by the 2026 Stanford AIMS Predictive AI Evaluation Competition. Our proposed approach builds on existing work by training a model built from two neural networks that estimate a system’s ability and a question’s difficulty using text embeddings of the provided database of system information and prompts. Our approach will be evaluated by the competition hosts on a set of unseen tasks using the Brier ALC score. We expect our final product to outperform a standard baseline neural network that we will initially develop. Additionally, we hope to use test-time adaptation to further improve performance based on a small number of real outcomes provided by the competition test harness at test time.

I. INTRODUCTION

Many organizations, including schools, hospitals, and businesses, increasingly need to assess the suitability of AI systems for their intended use cases, but lack the extensive resources currently required. Our project has the goal of designing a reliable evaluator that can be used to predict the probability that a given AI system answers a specific question or task correctly, using only descriptions of the system’s properties, the question to be answered, and optionally several sample question-response pairs generated by the system, all without explicitly running the AI system on the target question. The 2026 Stanford AIMS Predictive AI Evaluation Competition provides data, a concrete objective, an evaluation service, and a leaderboard to further motivate this project, and evaluates predictive capacity by testing submitted models on held-out benchmarks [1].

AIMS competition organizer Sang Truong has published multiple papers motivating the topic. One such paper describes how public benchmarks play a significant role in steering LLM development, and highlights the issue that fundamental information limitations mean evaluators are only able to cover a subset of socially relevant capabilities with their evaluation tasks [2]. The authors propose randomized evaluation mechanisms as a way to incentivize model developers to train to cover a broader set of socially relevant tasks. Our project complements this by developing a system that can evaluate on unknown inputs. Another paper compares and contrasts student evaluation and AI evaluation and provides an overview

on the struggles of reliably measuring capability change over time [3]. It proposes a roadmap that connects strengths through a unified framework organized around static measurement (where ability is inferred at a fixed point in time) and dynamic measurement (where ability is inferred as a trajectory). The main goal is stated to be to accelerate progress on the common problem of evaluating intelligent systems. Our project relates to this idea by taking Item Response Theory, a tool typically used to evaluate student test-takers, and applying it to AI models, using benchmarks as the test banks.

II. RELATED WORK

Zhou et al. introduce PSN-IRT, a neural Item Response Theory framework for analyzing large language model benchmarks. It uses separate feed-forward networks to estimate a model’s scalar ability and a question’s difficulty, discrimination, guessing rate, and feasibility. The authors use these estimates to examine benchmark quality and construct smaller evaluation sets [4].

Zintgraf et al. propose CAVIA, a meta-learning method that separates shared model parameters from a small set of task-specific context parameters. During meta-training, the shared parameters are optimized so that the context parameters can be updated from only a few examples, while at test time only the context parameters are changed. This provides a useful foundation to leverage test-time adaptation stage because it limits adaptation to a small ability correction instead of updating the entire predictor [5].

III. PROPOSED APPROACHES

We propose a deep neural predictor architecture that estimates the probability that an AI system, under a particular configuration, will successfully answer a question. We will implement three approaches with each adding to the previous design.

Approach 1 will be a basic binary classifier. Encoded system metadata and question features will be combined and passed through a feed-forward neural network (FNN), with a single sigmoid output representing the probability of success. The network will be trained using binary cross-entropy and backpropagation.

Approach 2 will adapt PSN-IRT using separate subject and item networks as developed in [4]. However, the subject network will produce an ability vector instead of a scalar,

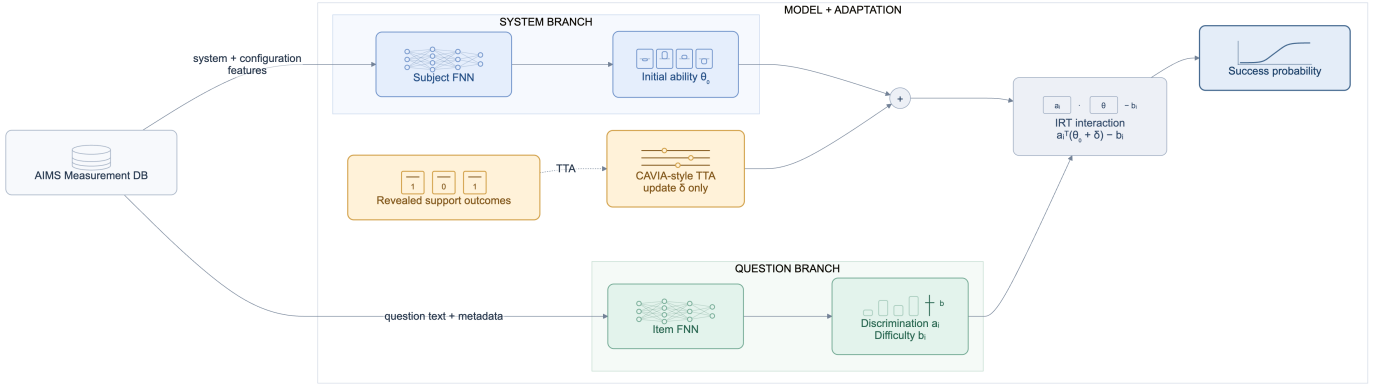


Fig. 1. Proposed system architecture.

while the item network will produce a discrimination vector and difficulty offset. Their interaction will lastly pass through a sigmoid activation to predict success. We will use system metadata and question representations instead of identifier-only inputs to support prediction on unseen questions, and omit the original guessing and feasibility parameters to simplify estimation. Both networks will be trained jointly using binary cross-entropy. The ability dimensions will be learned latent factors rather than predefined skills.

Approach 3 will add CAVIA-style meta-learning and test-time adaptation to Approach 2. During offline training, each episode will contain separate support and query questions for a subject–benchmark task. Support outcomes will update a small ability correction vector, while query loss after adaptation will train the shared networks through that update. At test time, the networks will remain fixed and only the correction will change using revealed support outcomes. This will test whether training specifically for adaptation improves predictions beyond the base predictor.

IV. SYSTEM DESIGN

A. Architecture

Fig. 1 shows the complete system with CAVIA-style adaptation. System metadata will be encoded into subject features, while question text will be embedded using a frozen pretrained text encoder and combined with permitted item metadata. Text representations will be cached, with long prompts processed in chunks. The subject FNN will produce the initial ability vector θ_0 , and the item FNN will produce discrimination a_i and difficulty b_i . The final prediction will be:

$$P(\text{correct}) = \text{sigmoid}[a_i^T(\theta_0 + \delta) - b_i]$$

The correction δ will start at zero and will be updated from revealed outcomes during adaptation. It will reset for each new subject–benchmark evaluation. Network sizes and other hyperparameters will be optimized using validation data.

B. Languages and Software Tools

Python will be the primary implementation language. PyTorch will be used to construct networks and perform gradient-

based adaptation, pandas and NumPy for data preparation, and SBERT for pretrained text representations. Scikit-learn will support pre-processing and evaluation, and Matplotlib will visualize learning curves and calibration plots. Git, model checkpointing, split manifests, and random seeds will support reproducibility.

C. Data Sets

The project will use the AI Measurement Data Bank, the public training dataset for the 2026 Stanford Predictive AI Evaluation Competition [1]. The current release contains six benchmarks, 12,632 items, 571,921 responses, and 543,735 execution traces. An item represents a question or task, a subject represents an AI system with a specific model and evaluation configuration, and a response records the outcome of an attempt. The proposed method will estimate the probability that a subject successfully completes an item using the subject and item information available through the competition interface. Execution traces will not be used as prediction inputs because they are unavailable at prediction time.

D. Evaluation

The proposed approaches will be compared against two baselines: a constant predictor that returns 0.5 (random guess) and the competition’s empirical-mean predictor, which uses revealed outcomes for the same subject and benchmark [6]. A subset of benchmarks will be held out during pre-competition testing to emulate the competition’s unseen-benchmark evaluation technique.

The primary evaluation metric will be Brier Area under Learning Curve (Brier ALC), the official competition metric [1]. In this case, Brier ALC is a weighted average of prediction error across test-time sample budgets of 0, 1, 3, 7, 15, and 31 samples. Lower scores indicate more accurate probability estimates across different amounts of test-time evidence. Expected Calibration Error will be reported to assess whether predicted probabilities match observed success rates. Model size, prediction time, and adaptation time will also be measured to assess the computational cost of the proposed approaches in order to meet secondary competition requirements. Additional

experiments will compare scalar and vector abilities, metadata-only and text-only inputs, and ordinary correction updates in Approach 2 against CAVIA-trained adaptation in Approach 3. We will also test temperature scaling fitted only on validation predictions. These comparisons will show whether the structured predictor and adaptation training each improve performance.

V. ACKNOWLEDGMENTS

OpenAI GPT-5.6 and Claude Opus 5 generative AI models were used to summarize references identified during the literature review. The proposed system architecture is a novel contribution of the authors.

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APPENDIX A

SCHEDULE AND TASK ASSIGNMENTS

Table I indicates each team member’s primary responsibilities. Table II assigns tasks for the entire 13-week project.

TABLE I
TEAM ROLES

Team member	Role
Dharaneesh	Data manager
Nick	Evaluation coordinator
Nishan	Architect
Steven	Team leader, lead writer

TABLE II
SCHEDULE AND TASK ASSIGNMENTS*

Week 1 (13-19 September 2026)	
Register team for competition	Steven
Analyze competition data sets	Dharaneesh
Research and draft abstract, introduction, related work	Nick
Research and draft proposed approaches and system design	Nishan
Project website	Steven
Finalize proposal	Steven
Week 2 (20-26 September)	
<i>Proposal due 22 Sep</i>	
Submit proposal package	Steven
Reserve long-lead services (compute, recording studio, etc.)	Steven
Exploratory data analysis	All
Split data into train/validate/test	Dharaneesh
Data pre-processing and retrieval infrastructure	Dharaneesh
Confirm access to cloud compute for training/testing	Nick
Clarify expectations for project website and visualizations	Steven
Design “Approach 1” on paper	All
Week 3 (27 September - 3 October)	
Review Approach 1 design with teaching staff	All
Implement, train, and internally test Approach 1	All
Week 4 (4-10 October)	
Submit Approach 1 to competition and analyze feedback	All
Document Approach 1 results for Checkpoint I report	All
Design “Approach 2” iteration on paper	All
Week 5 (11-17 October)	
Finalize Checkpoint I report	Steven
Review Approach 2 design with teaching staff	All
Implement, train, and test Approach 2	All
Week 6 (18-24 October)	
<i>Checkpoint I due 20 Oct (paper update)</i>	
Submit Checkpoint I report	Steven
Submit Approach 2 to competition and analyze feedback	All
Document Approach 2 results for Checkpoint II report	All
Design “Approach 3” iteration on paper	All
Week 7 (25-31 October)	
<i>Stanford AIMS Competition deadline 30 Oct</i>	
Finalize Checkpoint II report	Steven
Review Approach 3 design with teaching staff	All
Implement, train, and internally test Approach 3	All
Week 8 (1-7 November)	
Submit Approach 3 to competition and analyze feedback	All
Document Approach 3 results for final report	All
Week 9 (8-14 November)	
<i>Checkpoint II due 10 Nov (paper update)</i>	
Submit Checkpoint II report	Steven
Write concluding sections of final report	All
Week 10 (15-21 November)	
<i>Original AIMS Competition deadline; Eval site closes 20 Nov</i>	
Review conclusions with teaching staff	All
Finalize final report	Steven
Week 11 (22-28 November)	
<i>Thanksgiving Break</i>	
Write oral presentation slides * All	
Write 1-page summary	All
Week 12 (29 November - 5 December)	
Practice oral presentation	All
Record and edit video	All
Week 13 (6-11 December)	
<i>Final Presentations; Final Project Package due 11 Dec</i>	
Present oral presentation to class	All
Submit final project package	Steven

* In later weeks, tasks will be subdivided and assigned to individuals.